THE ISOTROPY LANDSCAPE ACROSS CUBICAL SITES WITH REVERSAL: RIGIDITY, CHARACTER REALIZABILITY, AND COVERING DESCENT

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Abstract. A companion paper showed that on each of the three standard cubical sites with reversal — De Morgan, Kleene, Boolean — the cubical-type and test model structures are separated by the same explicit map and the same explicit finite object. This paper determines how the *isotropy classification* behaves across the three sites, and finds that the conclusions coincide while the proofs cannot. Three results organize the phenomenon. First, a *rigidity discriminant*: an invariant interior cell of a two-stratum witness machine with twist δ can be slid to an end at stage zero if and only if the character $\varepsilon \circ \delta$ vanishes, where ε is the strata-swap character; genuinely twisted cells are permanent — they survive every stage of fibrant replacement, detected by a strict character projection to the interval quotient ($\mathbb{Z}/2$-cohomology of the test side detecting a type-side obstruction). Second, a *realizability divide*: twisted cells exist over a site if and only if the relevant sign character is realizable by a twisted element of the free algebra; on the Boolean site every stabilizer-compatible character is realizable (by signed XOR), on the Kleene site likewise (the Kleene inequality removes precisely the mixed points that carry the De Morgan parity obstruction), while the De Morgan site is the exceptional one where realizability fails and the certificate method of the classification papers applies. Third, *covering descent*: separations transfer along towers — quotients by characters with reflection-free cosets — via a covering theory for uniform Kan fibrant replacements; this bypasses the broken certificates and completes the Boolean and Kleene classifications in dimensions ≤ 3 with the same conclusions as the De Morgan one. All finite claims are machine-verified.

1. Introduction

Let $\square_{\mathcal{V}}$ be one of the three standard cube categories with reversal — De Morgan (the site of CCHM cubical type theory [2]), Kleene, or Boolean — and let W_{type} and W_{test} be the weak equivalences of the cubical-type and test model structures on presheaves [3, 5, 7]. The phase-diagram paper [6] showed that the separation $\mathsf{W}_{\text{type}} \subsetneq \mathsf{W}_{\text{test}}$, first proved on the De Morgan

2020 *Mathematics Subject Classification.* 55U35; 18N40, 08B20, 06D30, 20F55.

Key words and phrases. cubical sets, cubical type theory, model structure, isotropy quotient, De Morgan algebra, Kleene algebra, Boolean algebra, character, covering space, hyperoctahedral group.

site by an explicit map and an explicit finite object [8], transfers uniformly to all three sites. On the De Morgan site the separation was refined to a complete *isotropy classification* in dimensions $n \leq 3$ [4]: for each of the 98 subgroups $H \leq B_3$ of the hyperoctahedral group, either the quotient $\square^n/H$ carries the same homotopy type in both structures (with an explicit reason: coverings, contractions, median retractions, products), or an explicit separating map is attached to H (certificates, block transfer, homological obstructions).

This paper answers the question: *what happens to the classification on the Kleene and Boolean sites?* The conclusions turn out to be identical — the same subgroups separate, by attached maps, on all three sites — but the mechanism of proof is forced to change, and the way it changes is governed by a clean piece of algebra. The results below were found by exhaustive machine experiments (all finite claims in this paper are machine-verified) and then proved; we state them for the two-stratum witness machines of [4], recalled in Section 2.

The rigidity discriminant. For a mixed subgroup H with two reflection strata, the separation proofs of [8, 4] rest on a *nullity* statement: every invariant interior cell of the witness machine J_H/H can be slid to an end. Invariant cells carry a *twist* $\delta \in \mathrm{Hom}(H, H)$, and the machine has a *strata-swap character* $\varepsilon : H \to \mathbb{Z}/2$ recording which elements exchange the two strata.

Theorem 1.1 (Rigidity discriminant; Theorem 4.1)**.** *An invariant interior cell with twist δ is stage-zero slide-reachable to an end if and only if $\psi := \varepsilon \circ \delta = 0$. Moreover, cells with $\psi \neq 0$ are* permanent*: their classes differ from every end class at every stage of the fibrant replacement.*

The permanence is proved by a mechanism of independent interest: the coordinate t of the witness machine defines a strict equivariant projection $q : J_H/H \to \mathbb{L}$ to the interval quotient, which is a $K(\mathbb{Z}/2, 1)$ in the test structure; a genuinely twisted cell classifies a nontrivial double cover, and since cylinder homotopies in the type structure are in particular test homotopies, the test-side H^1 obstructs type-side nullity at every stage. On the Boolean Klein machine the projection factors *strictly* through the collage: the exact Boolean identity

$$F(d, a, t) \oplus F(d, \neg a, t) \ = \ t$$

exhibits $q = t^* \circ \bar{\Phi}$, where $t^* = [x \oplus y]$ is the unique nontrivial strict map from the full quotient of the square to $\mathbb{L}$.

The realizability divide. Whether cells with $\psi \neq 0$ *exist* is a question about the free algebras of the variety: ψ must be realized by a twisted element, $T \circ \sigma_h = \neg^{\psi(h)} T$. Here the three sites diverge (Theorem 3.2):

- *Boolean*: every character vanishing on the vertex stabilizers is realizable — by signed XOR, via the identity $e_S \circ \sigma_h = \neg^{\chi_S(h)} e_{h(S)}$ for $e_S = \bigoplus_{i \in S} x_i$;
- *Kleene*: likewise realizable. The De Morgan parity obstructions live on the *mixed* middle points of the literal cube — those with both a $(0,0)$-pair and a $(1,1)$-pair — and the Kleene inequality removes exactly these points from the domain of the free algebras (the Berman–Mukaidono representation [1]);
- *De Morgan*: realizability fails in the cases of the classification — this is precisely the content of the certificates of [4] — so the De Morgan site is the *exceptional* one, and it is the site where the original nullity proofs happen to work.

Covering descent. On the Kleene and Boolean sites the permanent twisted cells destroy the nullity proofs for most of the separating subgroups. The classification is completed by a third mechanism. Call a *tower* for H a character $\chi : H \to \mathbb{Z}/2$ whose kernel N has a known attached separation and whose nontrivial coset is reflection-free, so that $H/N \cong \mathbb{Z}/2$ acts freely on $\square^n/N$ and on the witness machine. We prove a covering theory for uniform Kan replacements (Lemma 5.2: coverings extend uniquely along the algebraic fibrant replacement, because the boxes of the uniform structure have connected domains) and a descent theorem (Theorem 5.3): *if the free quotient of an equivariant map is a type weak equivalence, so is the map itself*; hence non-equivalence descends to free quotients, and the attached separation of N produces one for H. Machine-verified towers exist for every Boolean- and Kleene-rigid subgroup of the $n \le 3$ classification (Table 2), yielding:

Theorem 1.2 (Landscape transfer; Theorem 6.1)**.** *In dimensions $n \le 3$, the isotropy classification of* [4] *holds verbatim on the Kleene and Boolean sites: the same* 78 *subgroups agree, and the same* 20 *separate by attached maps. The proofs distribute over three mechanisms — certificates, covering descent, and the fixed-point arguments — according to the realizability divide.*

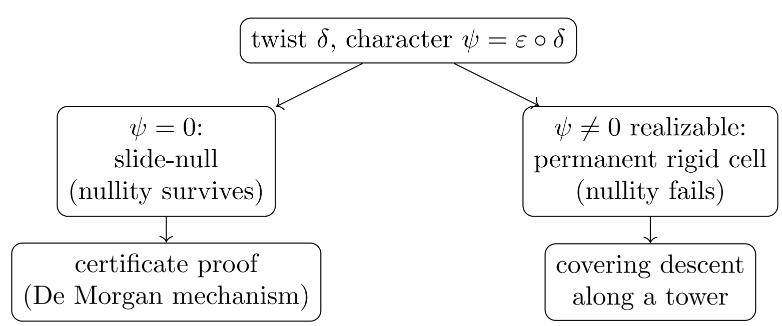

Figure 1. The discriminant routes each subgroup to a proof mechanism.

Relation to the companion papers. This paper is the third of a trilogy on the site-dependence of the type/test comparison: [6] determined which interval theories separate at all; the present paper determines how the classification's proofs deform across the separating sites; the underlying separation machinery is that of [8, 4]. The three mechanisms are all site-uniform — what varies is only which mechanism the realizability divide makes available.

2. Preliminaries

2.1. **Sites and free algebras.** We work over the De Morgan, Kleene, and Boolean cube categories; free algebras are represented as monotone functions on the literal cube $L_n = \{0,1\}^{2n}$ (De Morgan), monotone functions on the *unmixed* subposet $U_n \subseteq L_n$ obtained by removing the points that contain both a $(0,0)$-pair and a $(1,1)$-pair (Kleene; Berman–Mukaidono [1]), and arbitrary functions on $\{0,1\}^n$ (Boolean, negation the output complement). A point of L_n is *proper* when every pair is $(1,0)$ or $(0,1)$; proper points are unmixed, and the middle level of L_n intersects U_n exactly in the 2^n proper points. All sites carry the test and cubical-type model structures, and $\mathsf{W}_{\text{type}} \subseteq \mathsf{W}_{\text{test}}$ [8, 6].

2.2. **Witness machines, twists, and the swap character.** For a mixed two-stratum subgroup $H \le B_n$ (two reflections h_1, h_2 with disjoint fixed loci $\square^{m_1}, \square^{m_2}$), the *join model* J_H of [4] has coordinates $m_1 + m_2 + 1$ (the locus parameters and the join parameter t), a twisted H-action ν, and the median collage $\Phi = (F(\ell_1, \ell_2, t)\text{-components})$, $F(d,a,t) = (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a)$, descending to $\bar{\Phi}_H : J_H/H \to \square^n/H$. The *strata-swap character* $\varepsilon : H \to \mathbb{Z}/2$ takes value 1 on the elements exchanging the two loci; $\nu(h)$ acts on the t-coordinate by $t \mapsto \neg^{\varepsilon(h)} t$. An *invariant interior cell with twist* $\delta \in \mathrm{Hom}(H,H)$ is a cell c of J_H (interior: t-component nonconstant) with $c \circ \sigma_h = \nu(\delta(h)) \cdot c$ for all h; its class in (J_H/H) is invariant under the source substitutions, i.e., strict-map data for $\square^n/H \to J_H/H$ by strictification [8]. An invariant cell is *slide-null* when an invariant homotopy (in one further cylinder variable w) joins its class to a class supported in an end (t-component constant).

2.3. **Machine verification.** Throughout, "machine-verified" refers to the scripts accompanying the artifact (Section 7); on the Boolean site every parity system below is *exact* (a consistent assignment of truth-table bits is a cell), and on the Kleene site exactness is restored by a monotone-realizability filter over U_n.

3. The realizability divide

Lemma 3.1 (XOR character law)**.** *In the free Boolean algebra, for $S \subseteq \{1, \dots, n\}$ let $e_S := \bigoplus_{i \in S} x_i$. Then for every signed permutation $h = (\sigma, s)$,*

$$e_S \circ \sigma_h \;=\; \neg^{\chi_S(h)}\, e_{h(S)}, \qquad \chi_S(h) := \textstyle\sum_{i \in S} s_i \bmod 2.$$

In particular, if H stabilizes S then e_S realizes the sign character $\chi_S|_H$.

Proof. $e_S \circ \sigma_h = \bigoplus_{i\in S}(x_{\sigma(i)} \oplus s_i) = e_{\sigma(S)} \oplus \sum_{i\in S} s_i$, and $\oplus 1$ is the output complement. (Verified exhaustively for $n \leq 3$.) □

Theorem 3.2 (Realizability across the sites)**.** *Let $H \leq B_n$ and let $\psi : H \to \mathbb{Z}/2$ be a character. A ψ-twisted element ($T \circ \sigma_h = \neg^{\psi(h)} T$) of the free algebra on n generators exists:*

1. *over the Boolean site, if and only if ψ vanishes on the stabilizer of every vertex (and then a signed XOR realizes it);*
2. *over the Kleene site, in every case arising in the $n \leq 3$ classification in which the Boolean site realizes it (machine-verified: the full-U_3 parity systems with the monotone-realizability filter have exactly the same solvable twists as the Boolean systems);*
3. *over the De Morgan site, subject to the parity obstructions of* [4]*, which are supported on the mixed middle points of the literal cube; in the cases of the classification these obstructions are nonvacuous and realizability fails.*

Proof. (1) Necessity: at a vertex p fixed by h, twistedness forces $\psi(h) = 0$ pointwise. Sufficiency: pick orbit representatives of the vertex action and propagate values by the twisted equation; consistency around a stabilized point is exactly the vanishing hypothesis, and any function on $\{0,1\}^n$ is a free Boolean element. The XOR form follows from Lemma 3.1 when H stabilizes an S with $\chi_S = \psi$; in general the propagated solution is a signed XOR on each orbit. (2) is the machine statement; the structural reason is (3): the De Morgan obstruction systems, restricted to the unmixed poset, lose exactly the constraints supported on mixed points, and U_n contains no mixed middle point (its middle level is the 2^n proper points). The monotonicity of Kleene elements imposes no further obstruction in these cases — this is the nontrivial machine finding. (3) is the content of the certificate computations of [4], re-verified here. □

Remark 3.3. Theorem 3.2 inverts the naive expectation: adding equations to the variety (De Morgan $\to$ Kleene $\to$ Boolean) *increases* realizability, because the added equations destroy the carriers of the obstructions. The De Morgan site — the site of cubical type theory — is the exceptional site on which the certificate method works.

4. The rigidity discriminant

Theorem 4.1 (Discriminant)**.** *Let c be an invariant interior cell of J_H with twist δ, and $\psi := \varepsilon \circ \delta$.*

1. *If $\psi = 0$, then c is slide-null at stage zero: the slide $t \mapsto t \wedge \neg w$ in the cylinder variable, with the remaining coordinates extended w-constantly, is an invariant homotopy to the t=0 end.*
2. *If $\psi \neq 0$, then c is rigid at stage zero: no invariant homotopy joins its class to an end class.*

Proof. (1) The slide formulas are term operations available on all three sites; when $\psi = 0$ the deck action of the twist never flips t, so the slide is δ-equivariant, its $w{=}0$ face is c and its $w{=}1$ face has constant t-component. (2) Let $h \mapsto k_h$ assign to each generator a deck element witnessing the invariance of a homotopy c_4 with end faces. At the $w{=}1$ face the t-component is a constant ε_0; the twisted equation there reads $\varepsilon_0 = \psi'(h) \oplus \varepsilon_0$ for the homotopy's own character ψ', forcing $\psi' = 0$. Hence the $w{=}0$ face has an *untwisted* t-component; but that face is a deck translate of c, whose t-component is ψ-twisted (negation preserves twistedness). For h with $\psi(h) = 1$ and any point p, twistedness gives $T(\sigma_h p) = \neg T(-)$ while untwistedness gives equality — a pointwise contradiction, so no such function, hence no such homotopy, exists. □

Theorem 4.2 (Permanence)**.** *Suppose $\varepsilon \neq 0$ and let c be as in Theorem 4.1(2), with $g_c : \square^n/H \to J_H/H$ the corresponding strict map. Then the class of g_c in $[\square^n/H, R(J_H/H)]$ differs from the class of every end map, at every stage of the fibrant replacement.*

Proof. The t-coordinate is a strict map $J_H \to \square^1$, equivariant over the ε-action of H on $\square^1$ by reversal; since $\varepsilon \neq 0$ it descends to

$$q : J_H/H \longrightarrow \mathbb{L} := \square^1/\neg,$$

and both ends of J_H land on the unique vertex of $\mathbb{L}$, so q(end maps) are constant. Suppose $[g_c] = [\text{end}]$; composing with $R(q)$ gives $[q \circ g_c] = [\text{const}]$ in $[\square^n/H, R(\mathbb{L})]$. A type cylinder homotopy is in particular a test homotopy (the unit of the replacement is a type-trivial cofibration, hence a test equivalence), and $\mathbb{L}$ is a $K(\mathbb{Z}/2, 1)$ in the test structure on each site [8, 9], so

$$[\square^n/H, \mathbb{L}]_{\text{test}} \;\cong\; \operatorname{Hom}\bigl(\pi_1 \mathrm{el}(\square^n/H),\, \mathbb{Z}/2\bigr).$$

The map $q \circ g_c$ classifies the double cover with monodromy ψ: the twisted equation $T(\sigma_h\, p) = \psi(h) \oplus T(\cdots)$ is precisely the statement that transporting around the loop of h in the free part $BH \subseteq \mathrm{el}(\square^n/H)$ interchanges the sheets iff $\psi(h) = 1$. Since $\psi \neq 0$ the class is nontrivial, contradicting $[qg_c] = 0$. □

Remark 4.3. Theorem 4.2 is a type-side obstruction detected by test-side covering theory. On the Boolean Klein machine the projection q factors strictly through the collage: the identity $F(d, a, t) \oplus F(d, \neg a, t) = t$ (verified on all of $\mathrm{FB}(2)^3$) gives $q = t^* \circ \bar{\Phi}$ with $t^* = [x \oplus y]$ the unique nontrivial strict map $\square^2/B_2 \to \mathbb{L}$.

5. Covering descent

Throughout this section G is a finite group acting on a presheaf X *freely on nonempty cells*: no nontrivial element fixes a nonempty cell at any level. Write $Y := X/G$ (levelwise orbits); $p : X \to Y$ is then a levelwise principal G-covering.

Lemma 5.1 (Coverings are local systems)**.** *A presheaf map $E \to Z$ that is a levelwise principal G-covering is the same data as a functor* $\mathrm{el}(Z) \to G\text{-Tors}$*, i.e., a $\pi_1(\mathrm{el}Z)$-set with free transitive G-action. In particular isomorphism classes of G-coverings of Z are classified by* $\mathrm{Hom}(\pi_1\mathrm{el}(Z), G)$ *up to conjugacy, and cylinder-homotopic maps pull coverings back to isomorphic coverings.*

Proof. A presheaf over Z is a presheaf on $\mathrm{el}(Z)$; the levelwise torsor condition says each value is a G-torsor and each structure map a torsor map, which is a local system valued in G-torsors. The classification and homotopy invariance are the standard statements for local systems on a small category, via the nerve. $\square$

Lemma 5.2 (Unique extension of coverings)**.** *Let $Z \hookrightarrow Z'$ be a relative cell complex whose attachments are boxes of the uniform structure ($(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$-shaped, e.g. the algebraic fibrant replacement $Z' = R(Z)$ of* [8]*). Then every G-covering of Z extends to a G-covering of Z', uniquely up to unique isomorphism; and the total object of the extension of $p : X \to Y$ along $Y \hookrightarrow R(Y)$, written P, is a uniformly fibrant object containing X, with free G-action, $P/G \cong R(Y)$, and $X \hookrightarrow P$ a type-trivial cofibration. In particular P is a G-equivariant fibrant replacement of X realizing the replacement of the quotient.*

Proof. The domain $\mathrm{dom}(j) = S \times \square^1 \cup \square^n \times \{\varepsilon\}$ of a box is *connected*: every cell of $S \times \square^1$ has an ε-face chain landing in the representable part, which has a top cell. A local system extends uniquely along the attachment of a cell with connected attaching data (the monodromy of the attachment is determined, and el of the extended object deformation retracts onto the old part together with the new cell's boundary data); iterating through the transfinite composition gives the unique extension, and its total object P is built from X by attaching, over each downstairs box, its $|G|$ disjoint lifts — disjoint because the preimage of a connected subobject under a covering splits into $\leq |G|$ copies, and exactly $|G|$ here since the box lifts (its monodromy is trivial, being defined over the contractible $\mathrm{dom}(j)$-shape). Hence $X \hookrightarrow P$ is a transfinite composition of pushouts of boxes, i.e., a type-trivial cofibration. Fibrancy of P: given a box over P, project to $R(Y)$, fill uniformly, and lift the filler matching the given lift at the ε-end generator; the lift agrees with the given data on all of $\mathrm{dom}(j)$ by connectedness, and strict naturality is preserved because restriction preserves the ε-end base cell. $P/G \cong R(Y)$ holds levelwise. $\square$

Theorem 5.3 (Covering descent)**.** *Let G act freely (on nonempty cells) on X and X', let $f : X \to X'$ be equivariant, and suppose the quotient map $\bar{f} : X/G \to X'/G$ is a type weak equivalence. Then f is a type weak equivalence. Contrapositively, non-equivalence descends: if $f \notin \mathsf{W}_{\mathrm{type}}$ then $\bar{f} \notin \mathsf{W}_{\mathrm{type}}$.*

Proof. Write $Y = X/G$, $Y' = X'/G$, and let P, P' be the equivariant replacements of Lemma 5.2. The functorial replacement $R(\bar{f}) : R(Y) \to R(Y')$ pulls the covering P' back to a covering of $R(Y)$ which restricts on Y to $\bar{f}^*(X' \to Y') \cong (X \to Y)$ (equivariance of f); by the uniqueness of Lemma 5.2 this pullback is P, giving a canonical equivariant $F : P \to P'$ over $R(\bar{f})$ with $F|_X = f$ up to the identification. Since $\bar{f} \in \mathsf{W}_{\mathrm{type}}$ and all objects are cofibrant with $R(Y), R(Y')$ fibrant, $R(\bar{f})$ has a homotopy inverse $\bar{g}$ with cylinder homotopies $\bar{H} : \bar{g}R(\bar{f}) \simeq \mathrm{id}$ and $\bar{K} : R(\bar{f})\bar{g} \simeq \mathrm{id}$. By Lemma 5.1 the coverings are classified by characters of $\pi_1\mathrm{el}$, homotopy-invariantly; from $R(\bar{f})\bar{g} \simeq \mathrm{id}$ we get $\bar{g}^*[P] = \bar{g}^*R(\bar{f})^*[P'] = [P']$, so $\bar{g}$ lifts to an equivariant $\tilde{g} : P' \to P$. Similarly the homotopies lift (a cylinder $Z \times \square^1$ is connected when Z is, so extensions of coverings along cylinders are again unique), each lift agreeing with the expected end maps up to a deck translation. Thus $\tilde{g}F \simeq \gamma_a$ and $F\tilde{g} \simeq \gamma_b$ for deck automorphisms γ_a, γ_b; as these are isomorphisms, F has a left and a right homotopy inverse, hence is a homotopy equivalence. Since $X \hookrightarrow P$ and $X' \hookrightarrow P'$ are type-trivial cofibrations and F extends f, two-out-of-three gives $f \in \mathsf{W}_{\mathrm{type}}$. $\square$

Remark 5.4. The test-side counterpart is classical: el of a free quotient is the quotient of el, and free quotients of equivariant test equivalences are test equivalences (the Borel argument of [8]). Theorem 5.3 is its type-side partner, and the two together transfer an attached separation ($\bar{\Phi}_N \in \mathsf{W}_{\mathrm{test}} \setminus \mathsf{W}_{\mathrm{type}}$) from N to H along any tower.

6. The landscape in dimensions ≤ 3

6.1. **Machine census.** Table 1 records the exact stage-zero censuses.

	De Morgan	Kleene	Boolean
$\lvert F(2)\rvert$	168	84	16
essential invariant classes, $(J_{B_2}/B_2)([2])$	0	18	2
survivors: four-lines / B_3 / H_8 / H_{24}	0/6/5/7	9/19/5/7	9/19/5/7
sign characters realizable	no	yes	yes

Table 1. The realizability divide in the machine data (De Morgan values from [4]). "Survivors" are consistent nontrivial twists of the beyond-two-stratum certificate systems; on the Kleene side the counts are exact (monotone-realizability filter over U_3).

6.2. **Towers.**

Theorem 6.1 (Landscape transfer)**.** *On the Kleene and Boolean sites, in dimensions $n \leq 3$: every subgroup of the agreement families of* [4] *agrees (the median, product, covering, and fixed-point arguments are term-uniform or simplify — on the Boolean site the vertex-anchored contraction is the*

subgroup(s)	tower character	kernel (base of descent)
$K \times \mathbb{Z}_2$-blocks	—	(certificates apply)
B_2-block products	χ_s on the block	K-block ($\bar{\Phi}_K$, [6])
H_8 and conjugates	slot sign s_{fs}	K-block $\times 1$
four-lines 24	(unique free-coset char.)	A_4-type (certified)
H_{24}	total sign χ_{123}	A_4-type (certified)
B_3	(no free-coset tower)	v2/v3 arguments of [4]

TABLE 2. Machine-verified towers: in each row the coset of the character is reflection-free, so Theorem 5.3 applies over the named base.

single multiplexer formula [6]*); and every subgroup of the separating families carries an attached separating map:*

(1) *for the subgroups whose Boolean/Kleene certificate systems are consistent only for $\psi = 0$ twists (A_4-type, nested-*24*, the $|H| = 4$ two-stratum subgroups, the K-block products), the certificate argument of* [4] *transfers, with Theorem 4.1(1) supplying the slide;*
(2) *for the subgroups with permanent rigid cells, a machine-verified tower (Table 2) exists, and Theorem 5.3 descends the attached separation of its base;*
(3) *for B_3, the reflection-image survivors are handled by the fixed-vertex contraction ($\square^3/\langle\tau\rangle$ is contractible on every site: reflections fix vertices, and the interpolations of* [4, 6] *apply) and the free-image survivors by the cohomological argument of* [4]*, whose input — the $\mathbb{F}_2$-cohomology of* el *— is site-invariant.*

Hence the classification table of [4] *holds verbatim on all three sites.*

Proof. The distribution of subgroups over (1)–(3) is the machine content (Tables 1, 2); each mechanism has been established above or is quoted: (1) the certificate systems are exact on the Boolean site and exact-with-monotonicity on the Kleene site, so consistency-only-at-ψ=0 plus Theorem 4.1(1) reruns the nullity induction of [8, 4], whose remaining machinery is term-uniform [6]; (2) is Theorem 5.3 plus the base separations ((1) and [6]); (3) site-invariance of el follows from the stabilizer stratification, which depends only on the signed-cycle combinatorics of H (no self-negated elements on any site). $\square$

Remark 6.2 (What is genuinely different)**.** The classification's *conclusions* coincide across the three sites; what differs — provably — is the space of proofs. On the Kleene and Boolean sites the nullity route is *impossible* for the subgroups of Table 2 (Theorems 4.1 and 4.2), while on the De Morgan site the descent route is unnecessary. The permanent rigid classes are new homotopical objects of the Kleene and Boolean sites (their t-components are twisted functions with no De Morgan counterpart), and the question of the full homotopy type of the quotients they inhabit — e.g. whether $\square^2/B_2$ has the same homotopy type in both structures on the Boolean site — remains open, reduced by the free tower $\square^2/B_2 = W/\langle\bar{r}\rangle$ to an equivariant question about the Klein quotient.

7. Machine cross-checks

The scripts, archived in the artifact repository of this series

`https://github.com/r-shibusawa/cubical-strict-j`

(release `v1.18.0`, commit `ac39de6`, archived at DOI 10.5281/zenodo.21990464), verify: the XOR law (Lemma 3.1; $n \leq 3$); the Boolean and Kleene certificate systems for the two-stratum family (twelve subgroups of order ≥ 8: nine with permanent rigid cells, three slide-null, matching the discriminant $\varepsilon \circ \delta$ on every consistent twist — 12/12) and for the six beyond-two-stratum groups (Table 1); the interiority refinement (end-only twists are harmless); the slide-null decisions by exact parity search with reconstruction and verification of every positive; the Kleene monotone-realizability filter; the stage-zero censuses ((J/B_2): 0/18/2 essential classes across the sites); the towers of Table 2 (kernels, coset freeness, residual identifications); the freeness of the residual actions; and the collage factorization identity $F(d,a,t) \oplus F(d,\neg a,t) = t$.

Data availability. The verification scripts are openly available in the repository above, archived on Zenodo: release `v1.18.0`, commit `ac39de6`, DOI 10.5281/zenodo.21990464.

Use of AI tools. An AI assistant was used in the preparation of this manuscript and its verification scripts; the author takes full responsibility for the content.

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